

DSA0602 – Data Handling and Visualization

Assessment Test 2 (AT2) – CO1

VISHAAL M S

Register Number: 192424420

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Question 1

Nonlinear Axes [Python]

Code

```
import matplotlib.pyplot as plt

week = [1, 2, 3, 4, 5, 6, 7]
downloads = [50, 95, 180, 400, 820, 1600, 3100]

# 1. Linear y-axis
plt.figure(figsize=(7, 5))
plt.plot(week, downloads, marker='o', color='steelblue')
plt.title("App Downloads per Week (Linear Y-axis)")
plt.xlabel("Week"); plt.ylabel("Downloads")
plt.tight_layout()
plt.show()

# 2. Logarithmic y-axis
plt.figure(figsize=(7, 5))
plt.plot(week, downloads, marker='o', color='darkorange')
plt.yscale('log')
plt.title("App Downloads per Week (Logarithmic Y-axis)")
plt.xlabel("Week"); plt.ylabel("Downloads (log scale)")
plt.tight_layout()
plt.show()

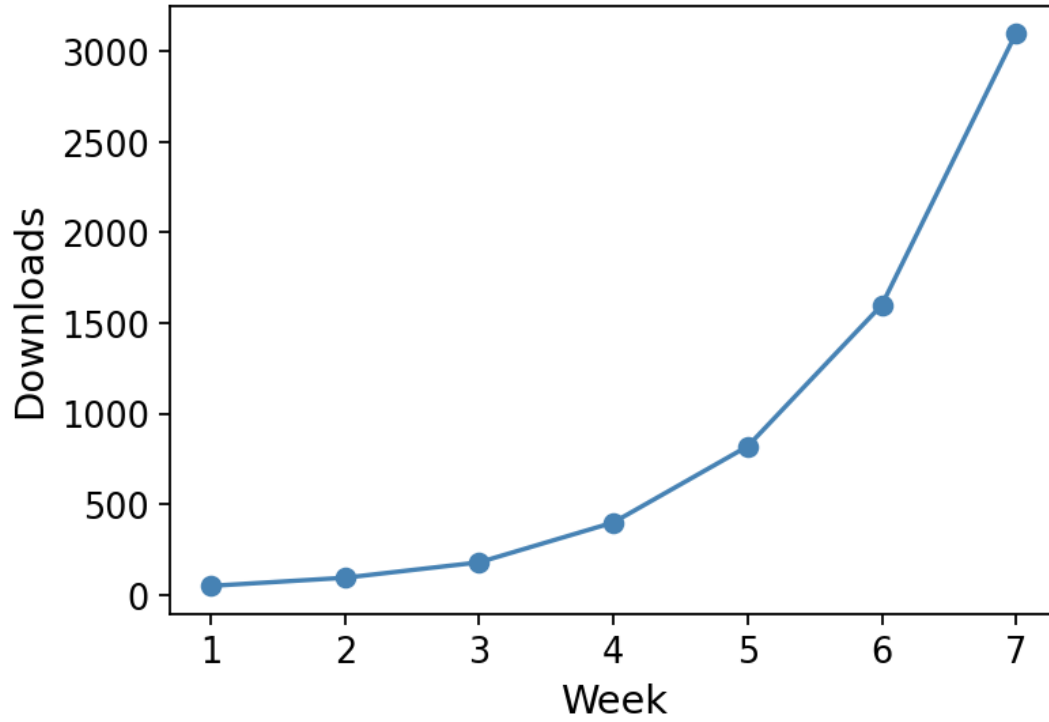
"""
3. Comparison:
On the linear axis, weeks 1-4 look almost flat and all the visual change is
crammed into weeks 6-7, because the later values (1600, 3100) are so much
larger than the early ones (50, 95). On the log axis, the growth is spread
evenly across the whole chart, and the line looks close to straight -
revealing that downloads are growing at a roughly constant MULTIPLICATIVE
rate (roughly doubling each week) rather than a constant additive amount.

4. Why log scale suits this dataset:
The data spans more than an order of magnitude (50 to 3100, ~62x) and grows
in a pattern that looks exponential/multiplicative (each value is roughly
~1.9-2x the previous one). Log scales are built for exactly this: they turn
exponential growth into a straight line and let small early-week values
remain visible instead of being squashed near zero.

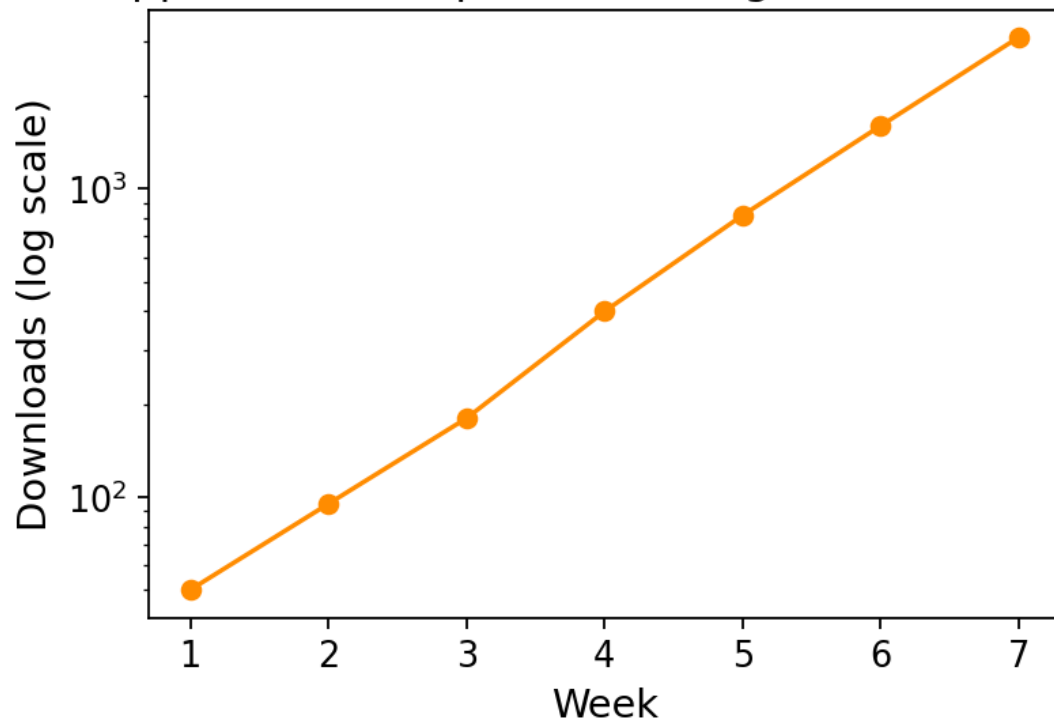
5. A situation where log scale should NOT be used:
When the data includes zero or negative values (e.g. profit/loss figures,
or temperature in Celsius) - log of zero or a negative number is undefined,
so the scale cannot represent those points at all. It's also a poor choice
when the audience needs to read and compare exact additive differences
(e.g. "this week gained 500 more downloads than last week"), since a log
axis visually compresses large absolute differences and can mislead readers
who aren't used to interpreting log spacing.
"""
```

Output

App Downloads per Week (Linear Y-axis)



App Downloads per Week (Logarithmic Y-axis)



Question 2

Legends and Annotations [Python]

Code

```
import matplotlib.pyplot as plt
import pandas as pd

df = pd.DataFrame({
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'Company X': [410, 430, 460, 480, 520, 500],
    'Company Y': [380, 395, 410, 430, 460, 490],
    'Company Z': [450, 460, 440, 470, 510, 530]
})

# 1. Monthly stock price trend for all three companies, with legend
plt.figure(figsize=(8, 5.5))
for company in ['Company X', 'Company Y', 'Company Z']:
    plt.plot(df['Month'], df[company], marker='o', label=company)
plt.title("Monthly Stock Price Trend")
plt.xlabel("Month"); plt.ylabel("Price (\u20b9)")
plt.legend(title="Company", loc='upper left')

# 2. Annotate the highest price point for each company
for company in ['Company X', 'Company Y', 'Company Z']:
    max_idx = df[company].idxmax()
    max_month = df['Month'][max_idx]
    max_val = df[company][max_idx]
    plt.annotate(f"Max: \u20b9{max_val}",
                xy=(max_month, max_val),
                xytext=(0, 15), textcoords='offset points',
                ha='center', fontsize=8,
                arrowprops=dict(arrowstyle='->', color='gray'))

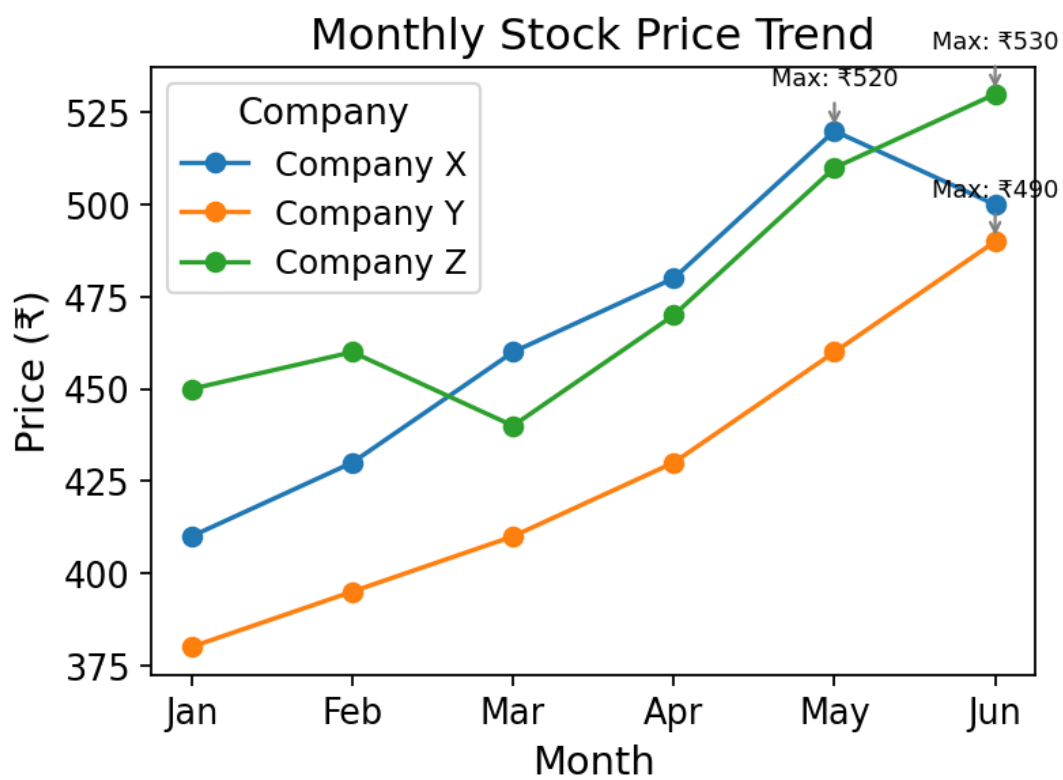
plt.tight_layout()
plt.show()

"""
3. Role of legends in multi-series visualizations:
A legend is the key that maps each visual encoding (here, line color) back
to what it represents (which company). Without it, a multi-line chart is
unreadable the moment more than one series is present, since color alone
carries no inherent meaning until it's labeled.

4. Best practices for legend placement:
Place the legend where it overlaps the least data (e.g. an empty corner of
the plot, or entirely outside the axes) rather than on top of a busy region
of the lines. Matplotlib's loc='best' (or manual inspection, as done above
with 'upper left') helps avoid obscuring data. For very crowded charts,
moving the legend outside the plot area (bbox_to_anchor) is safer than
forcing it inside.

5. An alternative to a legend - direct labeling:
Instead of a separate legend box, label each line directly at its
rightmost (or highest) point with a small text annotation naming the
series - e.g. writing "Company X" right next to the end of that line. This
removes the need for the reader's eye to jump back and forth between the
line and a separate legend key, and scales better when lines end at
clearly different heights.
"""
```

Output



Question 3

Mapping Data onto Aesthetics [R]

Code

```
library(ggplot2)

df <- data.frame(
  Item = c('I1', 'I2', 'I3', 'I4', 'I5', 'I6'),
  Orders = c(320, 540, 210, 690, 150, 410),
  Rating = c(4.2, 3.8, 4.6, 4.0, 4.8, 3.5),
  Category = c('Electronics', 'Grocery', 'Fashion', 'Grocery', 'Fashion', 'Electronics')
)

# 1 & 2. Scatter plot: Orders (x) vs Rating (y), color=Category, size=Orders
p <- ggplot(df, aes(x=Orders, y=Rating, color=Category, size=Orders)) +
  geom_point(alpha=0.8) +
  labs(title="Orders vs Rating by Category",
       x="Orders", y="Rating (out of 5)") +
  theme_minimal()
print(p)

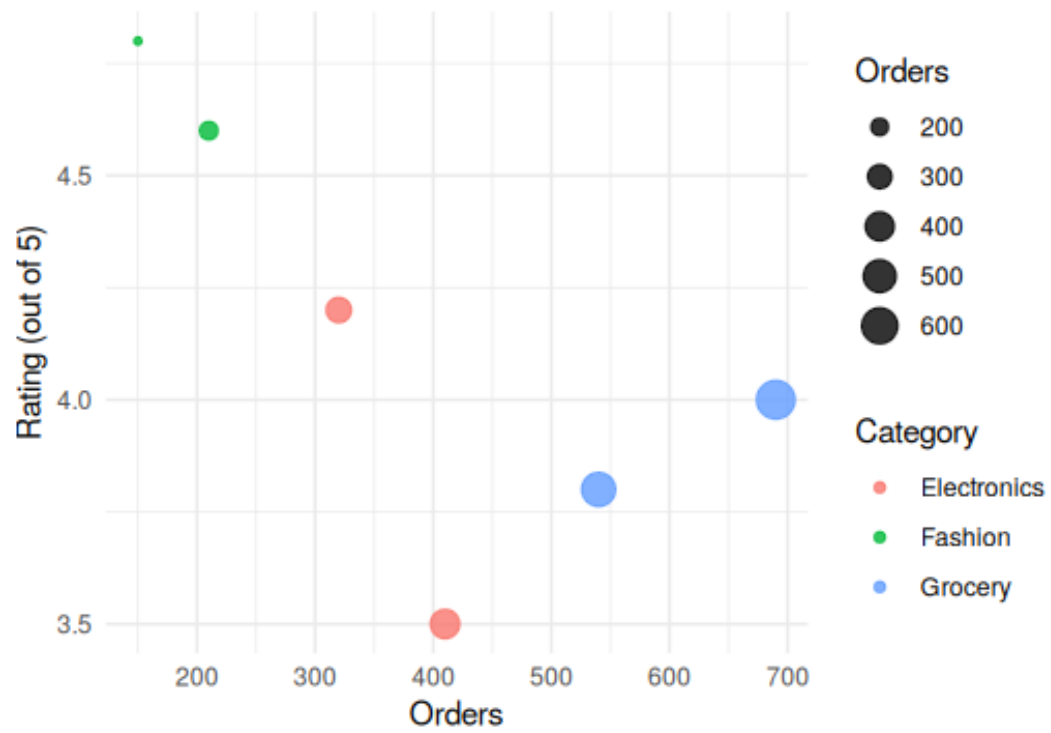
# 3. Interpretation:
# There is a mild negative relationship in this small sample: items with very
# high order counts (I4: 690 orders) tend to have middling ratings (4.0),
# while the highest-rated items (I5: 4.8, I3: 4.6) have comparatively low
# order counts (150, 210). This hints that popularity (orders) and perceived
# quality (rating) are not the same thing here - the most-ordered item is not
# the best-rated one. With only 6 points, this is a suggestive pattern rather
# than a statistically firm conclusion.

# 4. Scale types:
# - Orders: continuous (numeric count, meaningful magnitude and differences)
# - Rating: continuous (numeric, out of 5, though bounded)
# - Category: nominal (named groups - Electronics/Grocery/Fashion - with no
#   inherent order or ranking between them)

# 5. Alternative visualization:
# A grouped bar chart of average Rating per Category (or per Item) would
# better isolate the categorical comparison, since the scatter plot's size
# aesthetic (Orders) and the small sample size make it hard to compare
# categories directly. Alternatively, a bubble chart is essentially what's
# already built here - a faceted scatter plot (one panel per Category) could
# reduce visual clutter if the dataset grew larger.
```

Output

Orders vs Rating by Category



Question 4

Highlighting with Color [R]

Code

```
library(ggplot2)

df <- data.frame(
  ExpenseCategory = c('Rent', 'Groceries', 'Travel', 'Utilities', 'Entertainment'),
  MonthlySpend = c(15000, 8000, 4000, 3000, 2500)
)
df$Highlight <- ifelse(df$MonthlySpend == max(df$MonthlySpend), "Highest", "Other")
df$ExpenseCategory <- factor(df$ExpenseCategory, levels=df$ExpenseCategory[order(-df$MonthlySpend)])

# 1. Bar chart highlighting the highest-spending category
p <- ggplot(df, aes(x=ExpenseCategory, y=MonthlySpend, fill=Highlight)) +
  geom_col() +
  scale_fill_manual(values=c(Highest="firebrick", Other="grey70")) +
  labs(title="Monthly Expenses by Category (Highest Spend Highlighted)",
       x="Expense Category", y="Monthly Spend (Rs.)") +
  theme_minimal() +
  theme(legend.position="none")
print(p)

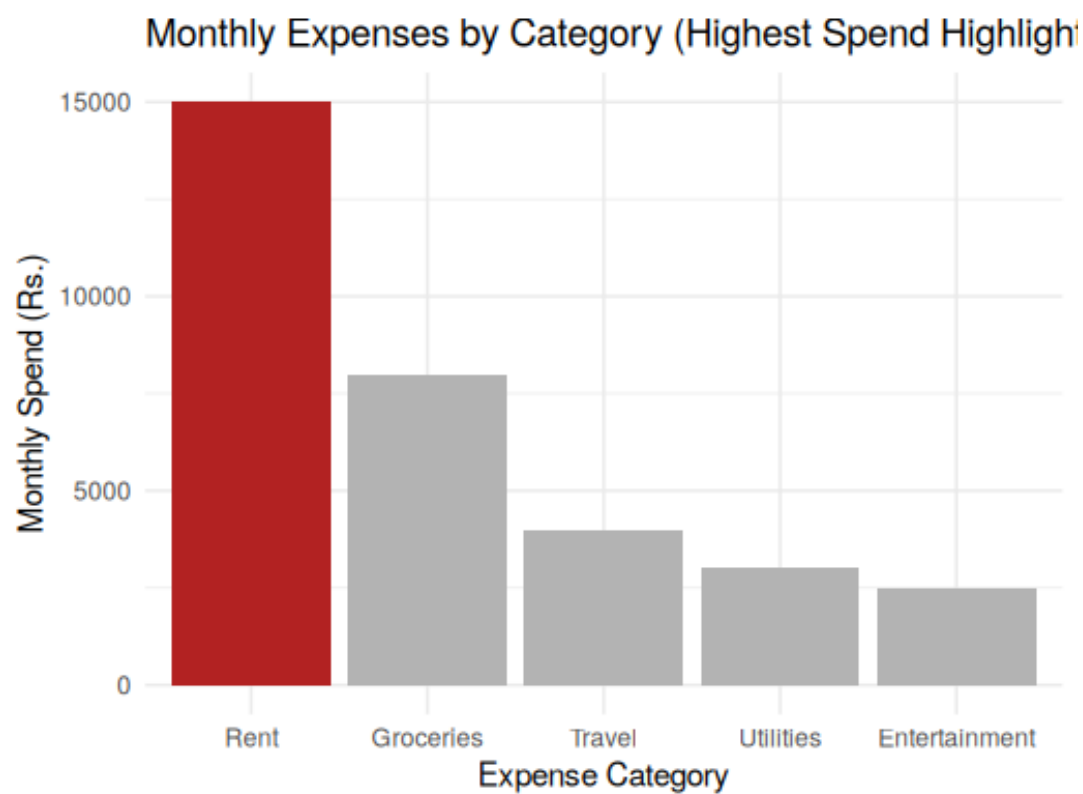
# 2. How highlighting improves interpretation:
# Highlighting draws the eye immediately to the single most important bar
# without requiring the reader to compare all five values manually. It
# exploits pre-attentive visual processing (color contrast is noticed almost
# instantly), turning a "read every number and compare" task into a
# "glance and see" task.

# 3. Highest-spending category: Rent, at Rs. 15,000/month - more than
# double the next-highest category (Groceries, Rs. 8,000).

# 4. Effect of excessive color use:
# If every bar (or too many categories) were given a distinct bright color,
# the highlighting effect is lost - the reader's eye has nothing to
# distinguish "important" from "normal" since everything competes for
# attention equally. Overuse of color also raises accessibility concerns
# (colorblind readers may struggle to distinguish many similar hues) and can
# make a chart look cluttered/unprofessional rather than informative.

# 5. Alternative highlighting technique (not color):
# Use annotation/labeling instead - e.g. adding a text label or an arrow
# pointing at the highest bar ("Highest: Rent"), or making that one bar's
# outline bold/thicker while keeping all bars the same fill color. Position
# (e.g. sorting the highlighted item first) or size (making it visually
# larger) are other non-color ways to draw attention.
```

Output



Question 5

Integrated Visualization [Python]

Code

```
import matplotlib.pyplot as plt
import pandas as pd

df = pd.DataFrame({
    'Country': ['C1', 'C2', 'C3', 'C4', 'C5'],
    'GDP': [12.5, 8.2, 21.0, 3.1, 5.6],
    'LiteracyRate': [92, 88, 95, 75, 80],
    'Population': [140, 65, 330, 210, 95],
    'Continent': ['Asia', 'Europe', 'N. America', 'S. America', 'Africa']
})

# 1 & 2. Bubble chart: x=GDP, y=Literacy Rate, size=Population, color=Continent
continents = df['Continent'].unique()
colors = plt.cm.Set2.colors

plt.figure(figsize=(8, 6))
for i, cont in enumerate(continents):
    sub = df[df['Continent'] == cont]
    plt.scatter(sub['GDP'], sub['LiteracyRate'], s=sub['Population'] * 3,
                color=colors[i % len(colors)], alpha=0.7, edgecolor='black', label=cont)
    for _, row in sub.iterrows():
        plt.annotate(row['Country'], (row['GDP'], row['LiteracyRate']),
                     textcoords="offset points", xytext=(0, 10), ha='center', fontsize=9)

plt.title("GDP vs Literacy Rate (bubble size = Population, color = Continent)")
plt.xlabel("GDP (\u20b9 Trillion)"); plt.ylabel("Literacy Rate (%)")
plt.legend(title="Continent", loc='lower right')
plt.tight_layout()
plt.show()

"""
3. Relationship between GDP and Literacy Rate:
There is a clear positive relationship in this sample - countries with
higher GDP tend to have higher literacy rates. C3 has both the highest GDP
(21.0 trillion) and the highest literacy rate (95%), while C4 has both the
lowest GDP (3.1 trillion) and the lowest literacy rate (75%). This is
consistent with the general real-world pattern that national wealth and
education access tend to move together, though 5 data points is too few to
draw a statistically robust conclusion.

4. Continuous and categorical variables:
- Continuous: GDP, Literacy Rate, Population (all numeric, meaningful
  magnitude and differences)
- Categorical: Continent (nominal - named groups with no inherent order),
  Country (nominal identifier/label, not really an analytic variable)

5. Recommended improvement:
Add a light trend line (e.g. a linear fit) through the bubbles to make the
positive GDP-literacy relationship visually explicit rather than something
the reader has to infer themselves. Additionally, using a log scale for
GDP could help if more countries with a wider wealth range were added,
since GDP values can vary by orders of magnitude across real countries.
"""
```

Output

GDP vs Literacy Rate (bubble size = Population, color = Continent)

